

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 08)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Explain how each of the following reduces the environmental impact of fishing.
[8 marks]

Square-mesh escape panels

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Minimum and maximum catchable sizes

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No-take zones

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Closed seasons

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01 . 2

Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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10

0 2 . 1

Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			ox
Use of materials			
Nature of wastes			
Temperature of processes			

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0 2 . 2

Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

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0 2 . 3

Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

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0 2 . 4

Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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10

0 3 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

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DNA databases and eDNA

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Image recognition software

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Acoustic monitoring / sonograms

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0 3 . 2

Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

10

04 . 1

Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

04 . 2

Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

10

0 5 . 1

Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

0 5 . 2

Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

0 5 . 3

Describe **three** threats to UK broadleaf woodland.

[3 marks]

1

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2

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3

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10

0 6 . 1

Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

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0 6 . 2

Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

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0 6 . 3

Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

06 . 4

Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

10

07 . 1

For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food

Fashion

Entertainment / pets

Furniture and ornaments

Traditional medicines

07 . 2

Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

1

2

3

0 7 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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10

0 8 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

- | | |
|---|--------------------------|
| A Close planting of a single non-indigenous species | ☐ |
| B Mixed indigenous species with a mixed age structure | ☐ |
| C Clear-felling on a 40-year rotation | ☐ |
| D Removal of dead wood | ☐ |

0 8 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

1

2

3

4

5

0 9 . 1 Explain how each of the following agricultural changes affects wildlife habitats.
[6 marks]

Drainage of wet grassland

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Application of fertiliser to a species-rich meadow

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Removal of hedgerows to enlarge fields

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0 9 . 2 Explain why some farmland habitats are valuable for wildlife conservation.

[2 marks]

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0	9	.	3
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Describe how agriculture contributes to climate change.

[2 marks]

10

10.1

Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS